



Cloud Computing

Assignment-Week 9

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

A factory requires machine-level anomaly detection within milliseconds, but it also needs to aggregate data from multiple machines for plant-wide analysis before sending summaries to the cloud. Which placement BEST matches the requirements?

- A) All processing in the centralized cloud
- B) Machine-level processing at the edge and plant-level processing at fog nodes
- C) All processing directly on individual sensors
- D) Plant-wide processing exclusively at the cloud

Correct Answer: B

Detailed Solution: Edge computing is appropriate for extremely latency-sensitive processing close to individual machines. Fog computing provides an intermediate layer capable of aggregating and analyzing information across multiple edge devices before forwarding relevant data to the cloud.

QUESTION 2:

A smart-city traffic system receives video streams from hundreds of cameras. Immediate object detection is performed near each camera, regional congestion patterns are analyzed at roadside micro-data centers, and long-term traffic trends are analyzed in the cloud. Which architecture BEST represents this system?

- A) Cloud-only architecture
- B) Fog-only architecture
- C) Edge–Fog–Cloud architecture
- D) Edge-only architecture

Correct Answer: C

Detailed Solution: The architecture uses Edge Computing for immediate processing near individual cameras, Fog Computing for regional aggregation and coordination, and Cloud Computing for large-scale historical analytics and storage. Therefore, the complete architecture is Edge–Fog–Cloud.



QUESTION 3:

An IoT system performs local filtering before forwarding data to the cloud. Which ordering BEST represents the typical amount of raw data transmitted toward centralized cloud infrastructure, from lowest to highest?

- A) Edge processing → Fog processing → Cloud-only processing
- B) Cloud-only processing → Fog processing → Edge processing
- C) Fog processing → Cloud-only processing → Edge processing
- D) Edge processing → Cloud-only processing → Fog processing

Correct Answer: A

Detailed Solution: When processing occurs at the edge, unnecessary data can be filtered before transmission. Fog processing can also aggregate data regionally. In a cloud-only approach, much more raw data may need to travel to centralized infrastructure.

QUESTION 4:

A fog node is running out of memory, but the infrastructure allows additional RAM and CPU resources to be assigned to that same node. Which scalability approach is being used?

- A) Horizontal scalability
- B) Temporal scalability
- C) Geographic scalability
- D) Vertical scalability

Correct Answer: D

Detailed Solution: Vertical scalability (scale-up) increases the resources of an existing node, such as CPU, RAM, or storage. Horizontal scalability, in contrast, adds additional nodes.

QUESTION 5:



Which statement BEST describes the division of responsibilities in a hierarchical Edge–Fog–Cloud architecture?

- A) Edge handles global optimization, while Cloud handles sensor-level decisions
- B) Edge handles immediate local actions, Fog coordinates regional resources, and Cloud performs global orchestration
- C) All three layers perform exactly the same tasks
- D) Only the Cloud is allowed to make decisions

Correct Answer: B

Detailed Solution: A hierarchical architecture distributes decision-making according to proximity and scope: Edge → immediate local decisions, Fog → regional coordination and resource management, Cloud → global policies, optimization, and long-term analytics

QUESTION 6:

A video analytics application requires its first-stage processing to occur within 10 ms of the camera, while its model-training component requires substantial computational resources. Which placement is MOST appropriate?

- A) Both components on a distant cloud server
- B) Both components on the smallest IoT device
- C) First-stage processing at the edge and model training in the cloud
- D) First-stage processing only in the cloud and training at the edge

Correct Answer: C

Detailed Solution: The latency-sensitive component should be placed close to the camera at the edge, while computationally intensive model training can be performed in the cloud, where greater computational resources are available

QUESTION 7:

A mobile device is moving rapidly between coverage areas of different edge servers while executing an offloaded task. Which issue becomes MOST critical?



- A) Static CPU allocation
- B) Service migration and task continuity during handover
- C) Cloud storage capacity
- D) VM image size only

Correct Answer: B

Detailed Solution: High mobility can cause a device to move from one edge server's coverage area to another. Maintaining service continuity may require task migration, service migration, or intelligent reassignment, while minimizing interruption and migration overhead.

QUESTION 8:

A cloud application requires additional resources because its primary CSP has insufficient capacity. A broker identifies a suitable secondary CSP and negotiates the required service conditions. Which cloud-federation function is being performed?

- A) Resource federation and inter-provider service provisioning
- B) VM deletion within a single CSP
- C) Local container orchestration
- D) Hardware virtualization

Correct Answer: A

Detailed Solution: In a cloud federation, a broker can discover resources from multiple CSPs and coordinate resource provisioning across providers. This enables an application to obtain additional capacity without depending entirely on a single provider.

QUESTION 9:

Why is trust management particularly important in a cloud federation?

- A) Resources and services may be obtained from independent providers with different security and reliability policies
- B) All CSPs are automatically controlled by the same organization
- C) Federation eliminates authentication requirements
- D) Trust is required only for physical servers



Correct Answer: A

Detailed Solution: Federated clouds involve independent CSPs, so providers may have different security practices, compliance requirements, reliability levels, and policies. Establishing trust and verifying provider behaviour is therefore critical.

QUESTION 10:

Which is the PRIMARY advantage of a Hybrid Cloud Federation for an organization handling sensitive workloads?

- A) It eliminates the need for security controls
- B) It forces all workloads into the public cloud
- C) It combines private-cloud control for sensitive workloads with public-cloud scalability
- D) It prevents workload migration between environments

Correct Answer: C

Detailed Solution: Hybrid federation provides a balance between security, control, compliance, and scalability. Sensitive applications can remain in private infrastructure, while less-sensitive or burst workloads can utilize public-cloud resources.